

Tactical EQS in True 3D

Environment queries that score real volume instead of a floor plane.

IN SHORT

LCM Nav3D ships EQS generators and tests that work in **true 3D**: candidate points anywhere in the navigable volume, scored by volumetric SVO line-of-sight, cover, threat-exposure, LCM Nav3D-path-distance and height. Author a query from these nodes exactly like stock EQS; the only differences are the `LCM Nav3D:`-prefixed generators/tests and that "distance" means *LCM Nav3D path*, not Euclidean or navmesh.

NOTE

Audience: anyone authoring tactical queries (sniper perch, flank, ambush, retreat). For perception-driven threat sources, pair with Tutorial 08.

The LCM Nav3D EQS node catalog

Generators (`UEnvQueryGenerator` , produce 3D points):

Node	Produces
LCM Nav3D: Points In SVO Volume	uniform 3D points in an AABB, filtered to unblocked SVO leaves
LCM Nav3D: Hemispherical Ring 3D	shells around a target (perch / orbit / high-ground); <i>Upper Hemisphere Only</i> option
LCM Nav3D: Points Along Path	points sampled along the LCM Nav3D path between two contexts
LCM Nav3D: Portal Proximity	points at macro-graph portal centres near a context (chokepoints)
<i>LCM Nav3D: Flow Field Directed</i>	<i>deferred: needs live flow-field state (see bottom)</i>

Tests (`UEnvQueryTest` , score/filter):

Node	Axis	Notes
LCM Nav3D: Line Of Sight 3D	visible to context (bool)	volumetric SVO LOS, not 2.5D
LCM Nav3D: Reachability 3D	an LCM Nav3D path exists (bool)	filter unreachable candidates
LCM Nav3D: Path Distance 3D	LCM Nav3D path length (float)	NOT Euclidean; Context = Querier
LCM Nav3D: Threat Exposure	summed SweepPenalty toward threats (float)	lower = safer
LCM Nav3D: Cover Score 3D	fraction of threats occluded-from (float)	higher = better cover
LCM Nav3D: Height Advantage	Z over context (float)	high ground
LCM Nav3D: Flow Field Alignment	<i>deferred: needs live flow-field state</i>	

Contexts (`UEnvQueryContext`):

Node	Returns
LCM Nav3D: Known Threats	actors tagged <code>ApexThreat</code>
LCM Nav3D: Threats From AI Perception	the querier's perceived threats (Tutorial 08)
LCM Nav3D: Active Flow Field	<i>deferred: names a live flow-field handle</i>

Step 1: Author a sniper-perch query

Create an EQS query `EQS_FindSniperPerch` :

- Generator:** LCM Nav3D: Hemispherical Ring 3D. Set `GenerateAround` = the threat context (Known Threats / Threats From AI Perception), `RadiusMin/Max` = your engagement band, `Upper Hemisphere Only` = true (perch above the threat).
- Test:** LCM Nav3D: Line Of Sight 3D (Context = threats). *Score, prefer should-have-LOS*: a perch must see its target.
- Test:** LCM Nav3D: Cover Score 3D (Context = threats). *Score↑*, because partial cover is good.
- Test:** LCM Nav3D: Height Advantage (Context = threats). *Score↑* to favour high ground.
- Test:** LCM Nav3D: Reachability 3D (Context = Querier). *Filter*: the agent must be able to get there.

The winning item is a reachable, high, in-cover position that still has line of sight to the threat: a sniper perch, chosen in full 3D.

Step 2: The EQS Testing Pawn

Drop an **EQS Testing Pawn** in the level, set its `QueryTemplate` to `EQS_FindSniperPerch`, and place threats so the contexts resolve. The candidate items render as colour-graded balls; tweak generator/test parameters and watch the scores update live.

NOTE

Use UE's stock `EQSTestingPawn` to author and preview these queries. The LCM Nav3D generators and tests render through it normally. Call **Build SVO In Editor** on the navigation manager first, or the preview has no navigation data to sample.

Step 3: Wire the result into a Behavior Tree

1. A `Run EQS Query` task/service runs `EQS_FindSniperPerch` and writes the best item to a blackboard `Vector` key.
2. A `MoveTo` (or the LCM Nav3D `FlyTo` BT task) moves the agent to that key.
3. Re-run on an interval (service) so the perch updates as threats move.

(StateTree: run the query from a task that stores the location, then transition to your LCM Nav3D move task; see [Tutorial 04](#).)

The five shipped example queries

Build these from the catalog above:

Query	Recipe	Status
<code>EQS_FindSniperPerch</code>	Hemispherical Ring 3D + Cover Score 3D + LOS 3D + Height Advantage	✓ buildable now
<code>EQS_FindFlankingApproach</code>	Points In SVO Volume + Threat Exposure↓ + Path Distance 3D	✓ buildable now
<code>EQS_FindAmbushCorner</code>	Portal Proximity + LOS 3D (prefer blocked) + Reachability 3D	✓ buildable now
<code>EQS_FindRallyPoint</code>	Flow Field Directed + Cover Score 3D	⌚ needs the deferred flow-field generator
<code>EQS_FindSafeRetreat</code>	Threat Exposure↑(inverted) + Path Distance 3D + Flow Field Alignment(inverted)	⌚ needs the deferred flow-field test

Deferred: the flow-field nodes

`LCM Nav3D: Flow Field Directed` (generator), `LCM Nav3D: Flow Field Alignment` (test), and `LCM Nav3D: Active Flow Field` (context) are **not yet shipped**. EQS generators/tests run **synchronously** inside a single query tick, but LCM Nav3D's cross-chunk flow field is **GPU-sampled asynchronously** (`SampleCrossChunkFlowField` is a batched async call). Wiring it into EQS needs a synchronous CPU-side flow-direction cache, which is follow-up work. The two example queries above that use them are blocked on that work; every other tactical query is fully available.

Related

- Tutorial 08: perception-driven threat sources (the auto-populated threat context).
- [Tutorial 04: The Same Agent in StateTree](#) (the move side).
- `ULcmTacticalQueryLibrary` : the same primitives as Blueprint nodes (`Is Segment LOS Clear` , `Find Best Cover` , `Find LCM Nav3D Path Sync` , ...).

Chapter 7 of the LCM Nav3D User Manual.

The complete manual is `LCM_Nav3D_Manual.pdf` , in this folder alongside every other document in the set. Start from `README.pdf` .

LCM Nav3D: True Volumetric 3D Navigation for Unreal Engine. Version 2.05 · Unreal Engine 5.8.

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